CS6140 Term Project Proposal: How Well Does Activity Recognition Work on New People?

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Problem

Phones and fitness trackers guess what you are doing (walking, sitting, climbing stairs) from motion sensors, and published results often report very high accuracy. My question is: how much of that accuracy holds up for a person the model has never seen? The data is cut into short windows that overlap by 50%. If windows are shuffled randomly into train and test sets, the test set contains near-copies of training windows from the same person, so accuracy may look better than it really is.

Data

UCI Human Activity Recognition dataset (Anguita et al., 2013), free and public. I have already downloaded and loaded it. It has 10,299 windows from 30 people, 6 activities, 561 ready-made features per window, and the raw sensor signals (9 channels × 128 samples). The official split puts 21 people in training (7,352 windows) and 9 in testing (2,947 windows), with no person in both. Classes are roughly balanced (1,406 to 1,944 windows each). Source: archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip

Plan

1. **Baseline:** logistic regression, SVM, random forest, and gradient boosting on the official person-wise split.
2. **Leakage test:** same data and models, random window split vs. split by person; measure the accuracy gap.
3. **Deep model:** a small 1D-CNN on the raw signals vs. models on the ready-made features.
4. **Error analysis:** confusion matrix (which activities get mixed up) and leave-one-person-out testing (how much accuracy changes from person to person).
5. **Scaling:** train on 2, 4, 8, 12, 16, and 21 people to see if adding people still helps. This shows what would change with 10x more data.

I will report accuracy and macro-F1 (each activity counts equally, so a model cannot look good just by doing well on common activities). The models go from simple to complex so I can see whether extra complexity actually helps.

Comparison with ground truth

The original paper reports 96% test accuracy using a tuned SVM on the official split. My first untuned SVM run gives 95.5%. I will tune my models using person-wise cross-validation and explain any remaining difference. The random-split vs. person-split comparison gives a second check, since both use the same data and models.

Feasibility

- **Data:** already downloaded and loaded (see Data).
- **Compute:** classical models run in minutes on a laptop or free Google Colab CPU; the CNN takes a few minutes on a free Colab GPU.
- **LLM use:** Claude for coding and debugging, under my existing account. The pipeline does not call any LLM, so no API credits are needed.
- **Code:** a working notebook loads the data and runs all five experiments.

Timeline and risks

By Nov 10 (checkpoint): baseline and leakage results, CNN training. By Dec 1: error analysis, scaling, and one extension (CNN under both splits, or a second dataset such as PAMAP2). By Dec 8: final report, written by me. If the leakage gap is small, that is still a useful result and I will explain why. If the CNN is unstable, the classical models stay the main result.